

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (EC) Examination

April 2015

EC201/EC201.01 Digital Electronics & Logic Design

Date: 22.04.2015, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Do as directed.

- (a) Convert $(105.15)_{10}$ into binary. [02]
- (b) Calculate: $(5531)_8 - (3261)_8 + (100)_{10} = (?)_{10}$ [02]
- (c) Determine the base x : $(16)_{10} = (100)_x$ [02]
- (d) Find out Gray code for binary number $(1010)_2$. [01]

Q - 2 Write any two answers. [14]

- (a) Implement $F(A,B,C,D) = \sum(0,1,3,4,8,9,15)$ using Multiplexer. [07]
- (b) Explain 4 bit binary ripple counter with count sequence. [07]
- (c) Explain PLA with block diagram. Also discuss PLA implementation with example. [07]

Q - 3 Write any two answers. [14]

- (a) Simplify following function using K-map. [07]
$$F(w,x,y,z) = \sum(0,1,2,4,5,6,8,9,12,13,14)$$
- (b) Simplify the following Boolean function by using the tabulation method. [07]
$$F(a,b,c,d) = \sum(0,1,2,8,10,11,14,15)$$
- (c) Reduce following function. [07]
$$F = A'B'C' + B'CD' + A'BCD' + AB'C'$$

SECTION – II

Q - 4 (a) Prove following Boolean Theorems. [04]

- (1) $X+X=X$ (2) $X+1=1$
- (3) $X+XY=X$ (4) $X(X+Y)=X$ (by duality)

- (b) Reduced following function and draw logic diagram. [03]

$$F = XY + X'Z + YZ$$

OR

- (b) Draw and Explain 4 bit ALU block diagram. [03]

Q – 5 Write any two answers. [14]

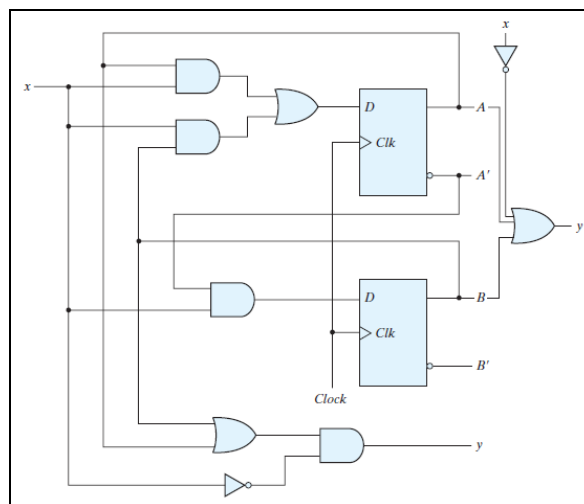
- (a) Explain De Morgan's Law with logic diagram and truth table. Apply De Morgan's Law and reduced following function. [07]

$$F = ((AB)' + A' + AB)'$$

- (b) Design Full Adder using two half adder and a OR gate with truth table and functions. [07]
 (c) Design BCD to Excess-3 code conversion logic diagram with truth table and functions. [07]

Q - 6 Write any two answers. [14]

- (a) Explain J-K flip flop with Logic diagram, graphic symbol, characteristic table and equation. [07]
 (b) Write state table for following clocked sequential circuit. Also draw state diagram for the circuit. [07]



- (c) Draw Logic diagram for following function with two input terminal gates. Design same Functions using $F = A(B + CD) + BC'$ [07]
 (i) Only NAND implementation
 (ii) Only NOR implementation
